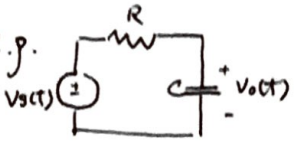
§14 Frequency Response

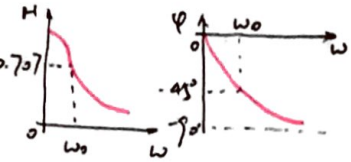
$H(\omega) = \frac{Y(\omega)}{X(\omega)}$ (transfer function = $\frac{\text{output phasor}}{\text{input phasor}}$)

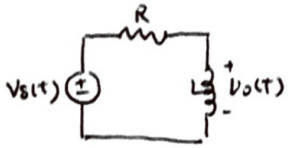
e.g.



$H(\omega) = \frac{1}{1+j\omega RC}$

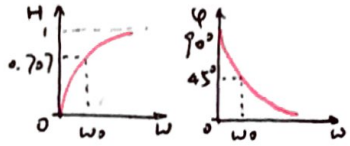
$\Rightarrow \begin{cases} H = \frac{1}{\sqrt{1+(\omega/\omega_0)^2}} \\ \varphi = -\arctan\frac{\omega}{\omega_0} \end{cases}$





$H(\omega) = \frac{j\omega L}{R+j\omega L}$

$\Rightarrow \begin{cases} H = \frac{\omega}{\sqrt{(1/\omega_0)^2+\omega^2}} \\ \varphi = \arctan\frac{1}{\omega_0\omega} \end{cases}$



Decibel Scale

power Gain $= \log_{10}\frac{P_2}{P_1}$

$G_{dB} = 10\log_{10}\frac{P_2}{P_1}$

$R_1 = R_2 \Rightarrow G_{dB} = 20\log_{10}\frac{V_2}{V_1}$

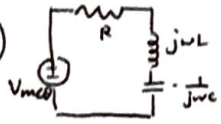
$= 20\log_{10}\frac{I_2}{I_1}$

$\bar{H} = H\angle\phi = He^{j\phi}$

$\ln\bar{H} = \ln H + j\phi$

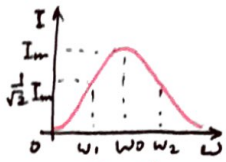
$H_{dB} = 20\log_{10}H$

Series Resonance



resonance $\Rightarrow Z = R + j\omega L + \frac{1}{j\omega C} = 0 \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$

- * LC short circuit
- * V, I in phase
- * $I = I_{max}$



bandwidth B

$\boxed{\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}}$

$\omega_1\omega_2 = \omega_0^2$

$\boxed{BW = \omega_2 - \omega_1 = \frac{R}{L}}$

$\boxed{Q = \frac{\omega_0}{BW}}$

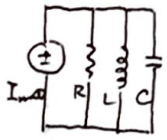
For high-Q circuit ($Q \geq 10$)

$\omega_{1,2} = \mp\frac{\omega_0}{2Q} + \omega_0 = \mp\frac{BW}{2} + \omega_0$

$\rightarrow Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$ $\quad BW = \frac{R}{L} = \frac{\omega_0}{Q}$

$V_L = V_C = QV_m$.

Parallel Resonance



resonance $\Rightarrow Y = \frac{1}{R} + j\omega C + \frac{1}{j\omega L} = 0 \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

- * LC open circuit
- * V, I in phase
- * $V = V_{max}$

$\boxed{\omega_{1,2} = \mp\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}}$

$\boxed{BW = \omega_2 - \omega_1 = \frac{1}{RC}}$ $\boxed{Q = \frac{\omega_0}{BW} = \omega_0 RC = \frac{R}{\omega_0 L}}$

Bode Plots

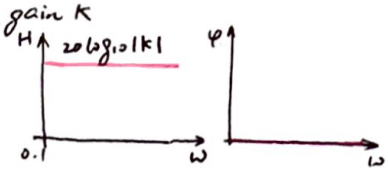
standard form

$$H(\omega) = \frac{K(j\omega)^{\pm 1}(1+j\omega/z_1)[1+j2\zeta_1\omega/\omega_k + (j\omega/\omega_k)^2]\cdots}{(1+j\omega/p_1)[1+j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]\cdots}$$

① constant term, the gain K

$H_{dB} = 20\log_{10}|K|$

$\varphi = \begin{cases} 0°, & K>0 \\ 180°, & K<0 \end{cases}$

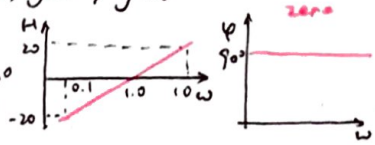


② pole/zero at the origin, $(j\omega)^{-1}$/$(j\omega)^1$

$(j\omega)$: $H_{dB} = 20\log_{10}\omega$, $\varphi = 90°$

$(j\omega)^{-1}$: $H_{dB} = -20\log_{10}\omega$, $\varphi = -90°$

$(j\omega)^N$: $H_{dB} = N\log_{10}\omega$, $\varphi = 90N°$



slope = 20dB/decade. (1~10)

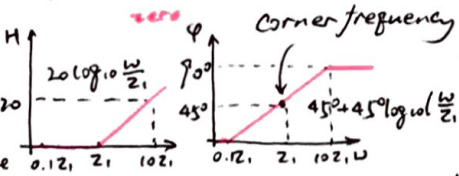
③ simple pole/zero

$(1+j\omega/z_1)$: $H_{dB} = 20\log_{10}|1+\frac{j\omega}{z_1}|$

zero $\quad \varphi = \arctan\frac{\omega}{z_1}$

$1/(1+j\omega/p_1)$: slope of H: -20dB/decade

pole $\quad$ slope of φ: -45°dB/decade

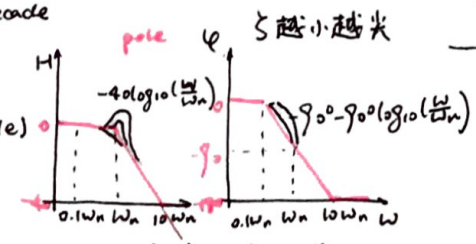


④ quadratic pole/zero

$1/[1+j2\zeta_2\omega/\omega_n + (j\omega/\omega_n)^2]$: (pole)

$H_{dB} = -20\log_{10}|\cdots|$

$\varphi = -\arctan\left(\frac{2\zeta_2\omega/\omega_n}{1-\omega^2/\omega_n^2}\right)$



zero is similar except

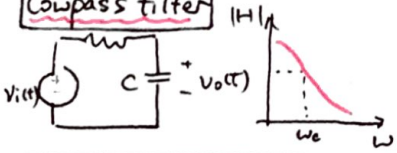
40dB/decade $\quad$ 90°/decade

Passive Filters

$H_{(\omega)} = \frac{V_o}{V_i} = \frac{1}{1+j\omega RC}$

$|H(\omega_c)| = \frac{1}{\sqrt{2}} \Rightarrow \omega_c = \frac{1}{RC}$

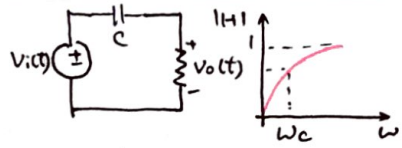
lowpass filter



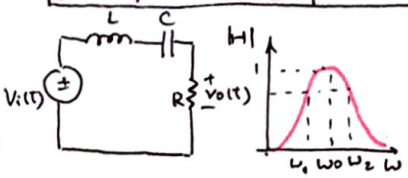
highpass filter

$H(\omega) = \frac{V_o}{V_i} = \frac{j\omega RC}{1+j\omega RC}$

$\omega_c = \frac{1}{RC}$



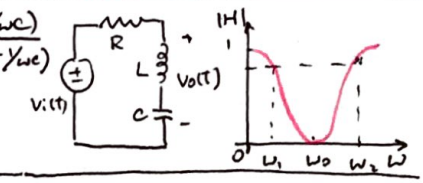
bandpass filter



$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R+j(\omega L - 1/\omega C)}$

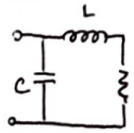
$\omega_0 = \frac{1}{\sqrt{LC}}$

bandstop filter

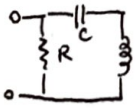


$H(\omega) = \frac{V_o}{V_i} = \frac{j(\omega L - 1/\omega C)}{R+j(\omega L - 1/\omega C)}$

$\omega_0 = \frac{1}{\sqrt{LC}}$



$\omega_0 = \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$



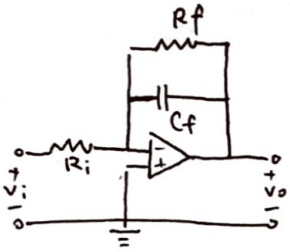
$\omega_0 = \frac{1}{\sqrt{LC}}$

Active Filters

first order low-pass filter

$H(\omega) = -\frac{R_f}{R_i}\frac{1}{1+j\omega/\omega_c}$

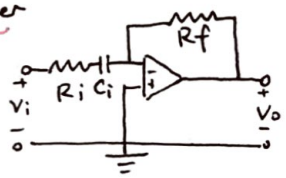
$\omega_c = 1/(R_f C_f)$



first order high-pass filter

$H(\omega) = -\frac{R_f}{R_i}\frac{j\omega/\omega_c}{1+j\omega/\omega_c}$

$\omega_c = 1/(R_i C_i)$

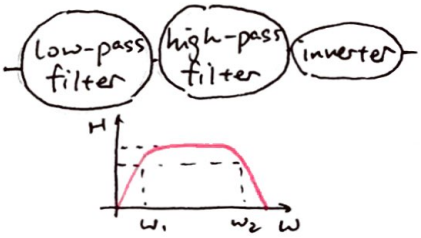


band-pass filter

H 相乘

$\omega_1 = 1/(RC_2)$ $\omega_2 = 1/(RC_1)$

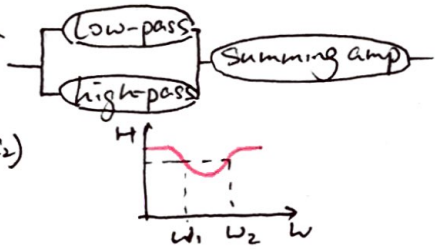
$\omega_0 = \sqrt{\omega_1\omega_2}$ $BW = \omega_2 - \omega_1$



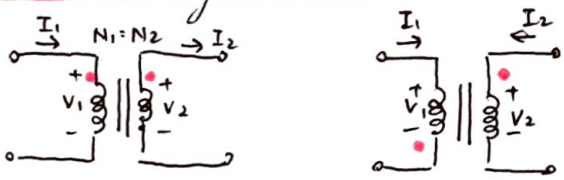
band-reject filter

H 相加

$\omega_1 = 1/(RC_1)$ $\omega_2 = 1/(RC_2)$



Ideal Transformers (k=0)



$\frac{V_2}{V_1} = \frac{N_2}{N_1}$ $\frac{I_2}{I_1} = \frac{N_1}{N_2}$ $\frac{V_2}{V_1} = -\frac{N_2}{N_1}$ $\frac{I_2}{I_1} = \frac{N_1}{N_2}$

$Z_{in} = \frac{1}{n^2} Z_L$ (reflected)

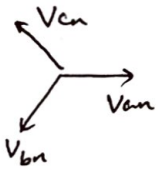
$V_{in} = \frac{V}{n}$

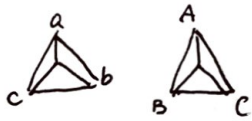
$I_{in} = nI$

§12 Three-Phase Circuit

$$\begin{cases} V_{an} = V_p\angle 0^\circ \\ V_{bn} = V_p\angle -120^\circ \\ V_{cn} = V_p\angle 120^\circ \end{cases}$$

positive





Y-Y $V_L = \sqrt{3}V_p$

$$\begin{cases} \tilde{V}_{ab} = \tilde{V}_{an}\angle 30^\circ \\ \tilde{V}_{bc} = \tilde{V}_{bn}\angle 30^\circ \\ \tilde{V}_{ca} = \tilde{V}_{cn}\angle 30^\circ \end{cases}$$

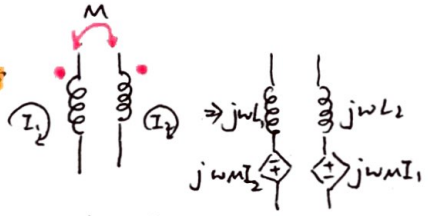
Y-Δ $V_L = \sqrt{3}V_p$

$$\begin{cases} \tilde{V}_{ab} = \tilde{V}_{an}\angle 30^\circ \\ \tilde{V}_{bc} = \tilde{V}_{bc}\angle 30^\circ \\ \tilde{V}_{ca} = \tilde{V}_{cn}\angle 30^\circ \end{cases}$$

$I_L = \sqrt{3}I_p$

$\tilde{I}_a = \tilde{I}_{AB}\sqrt{3}\angle -30^\circ$

§13 Magnetically Coupled Circuits



Self-inductance $v = L\frac{di}{dt}$

Mutual-inductance

in series $\begin{cases} L = L_1 + L_2 + 2M & \text{(aiding)} \\ L = L_1 + L_2 - 2M \end{cases}$

in parallel $L = \frac{L_1 + L_2 - M^2}{L_1 + L_2 - 2M}$ (opposing)

energy in a coupled circuit $w = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 + MI_1I_2$ (同进)

$k = \frac{M}{\sqrt{L_1L_2}}$ $(0 \le k \le 1)$

$w = \frac{1}{2}L_1I_1^2 + \frac{1}{2}L_2I_2^2 - MI_1I_2$ (一进一出)

Δ-Δ $I_L = \sqrt{3}I_p$

$\tilde{I}_a = \tilde{I}_{AB}\sqrt{3}\angle -30^\circ$

Δ-Y

line > voltage
phase

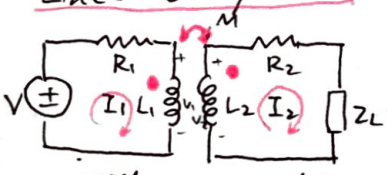
line > current
phase

$p(t) = 3V_pI_p\cos(\theta) = \text{const}$

$P_p = V_pI_p\cos\theta$ $Q_p = V_pI_p\sin\theta$

$S_p = P_p + jQ_p = \tilde{V}_p\tilde{I}_p^*$

Linear transformer



primary winding

secondary winding

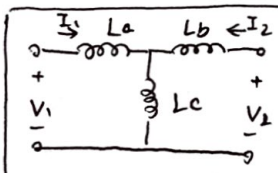
$$\begin{cases} V_1 = L_1\frac{di_1}{dt} - M\frac{di_2}{dt} \\ V_2 = M\frac{di_1}{dt} - L_2\frac{di_2}{dt} \end{cases}$$

$$\Rightarrow \begin{cases} \tilde{V}_1 = j\omega L_1\tilde{I}_1 - j\omega M\tilde{I}_2 \\ \tilde{V}_2 = j\omega M\tilde{I}_1 - j\omega L_2\tilde{I}_2 \end{cases}$$

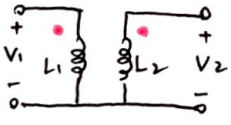
$Z_{in} = \frac{\tilde{V}_1}{\tilde{I}_1} = R_1 + j\omega L_1 + \frac{\omega^2M^2}{R_2 + j\omega L_2 + Z_L}$

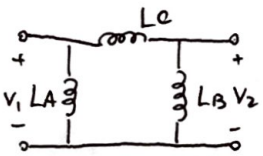
reflected impedience

(不受点影响)



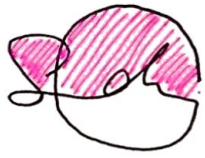
$$\begin{cases} L_a = L_1 - M \\ L_b = L_2 - M \\ L_c = L_3 - M \end{cases}$$





$$\begin{cases} L_A = \frac{L_1L_2 - M^2}{L_2 - M} \\ L_B = \frac{L_1L_2 - M^2}{L_1 - M} \\ L_C = \frac{L_1L_2 - M^2}{M} \end{cases}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix}\begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$



VE215 Final